Camera-Assisted Radar Detection Clustering for Extended Target Tracking

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***Abstract*— Radar measurements from extended targets, such as vehicles or pedestrians, need to be clustered before they can be tracked by extended target tracking algorithms. However, clustering is challenging in autonomous driving scenarios, where multiple objects of different shapes and sizes remain in close proximity. Traditional clustering algorithms often produce wrong clusters due to merging and splitting, which degrade the pose estimation and tracking performance. In this work, we propose an approach to improve radar clustering using additional camera input. Our approach consists of two steps: first, we propose a range-based adaptive parameter selection method for radar-only clustering, which adjusts the clustering parameters according to the distance of the radar measurements. Second, we propose a camera-assisted clustering algorithm, which uses the camera images to refine the radar clusters by exploiting the visual information of the objects. We evaluate our approach by performing extensive simulations using the publicly available nuScenes dataset, which contains synchronized radar and camera data. Simulation results indicate that the proposed approach significantly improves the clustering performance in terms of accuracy and robustness.**

***Index Terms*— Autonomous vehicle, camera data, clustering, extended target, radar data, sensor fusion.**

I. Introduction

AUTONOMOUS driving is ubiquitous in the future of mobility, which aims to enable vehicles to operate safely and efficiently without human intervention. Ensuring the safety of such autonomous vehicles requires them to have effective perception capabilities of their dynamic and uncertain environment. However, no single-modal sensor can provide complete and accurate information for all scenarios and conditions [1], [2]. Therefore, multimodal sensing, which combines different types of sensors, such as cameras, lidars, and radars, is essential for autonomous driving [3], [4]. Moreover, multimodal sensing can improve the robustness and resilience of autonomous driving by mitigating the limitations and failures of individual sensors [4], [5], [6], [7].

Radars are one of the most commonly used sensors in autonomous driving, as they can provide reliable and robust detection of various objects in different weather and lighting conditions [7], [8]. In modern radar systems, high-resolution sensors yield multiple measurements from various facets of objects with nonnegligible spatial extent [9], [10]. Consequently, an indispensable preprocessing step involves segregating measurements originating from distinct objects into distinct groups. Such partitioning is essential for subsequent tasks, such as data association and tracking. Clustering plays a pivotal role in partitioning radar measurements into coherent and distinct groups. Unlike supervised learning methods, clustering does not rely on the prior knowledge of cluster shapes or quantities [11], [12], [13]. However, in complex traffic scenarios, conventional clustering algorithms encounter many practical challenges.

One of the challenges of clustering radar data is dealing with closely spaced targets of various shapes and sizes. In these situations, the external density difference between adjacent clusters may be similar to that of the internal density difference within a single cluster, due to the presence of clutter and missed detections. Therefore, relying solely on spatial density information from radar data may result in inaccurate clustering. A common problem of applying traditional density-based clustering algorithms to radar data is that they may either split an extended target into multiple clusters (splitting error) or merge several closely spaced targets into one cluster (merging error) [14]. These errors introduce uncertainty in the cluster origin and affect the subsequent steps of pose estimation and tracking.

In our prior work in [15], a camera-assisted density-based clustering approach to address the aforementioned issue was proposed. This approach capitalizes on the categories and bounding boxes of targets extracted from images at each time step, enhancing the radar-only clustering process. Specifically, radar measurements are partitioned based on the overlap between their image projections and the bounding boxes. Subsequently, these measurements are clustered separately, with parameters that dynamically adapt to the range and class of the targets. Preliminary simulation results demonstrate the

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improved clustering accuracy. However, it is important to note that this approach faces limitations related to overlapping bounding boxes and coverage discrepancies, which hinder its effectiveness in mitigating merging and splitting errors.

In this work, we build upon existing research by tackling the challenges associated with overlapping bounding boxes and coverage discrepancies. In addition, we introduce a novel approach involving 2-D-assignment-based merging to mitigate errors in this extended framework. Our primary contributions to this study are summarized as follows.

1) *Enhancing Radar-Only Clustering With Adaptive Criteria:* Two distinct adaptive criteria to improve radar-only clustering are proposed. The first criterion focuses on adaptability concerning the radial and tangential density, while the second criterion specifically addresses class adaptability for short-range clustering.
2) *Bounding Box-Based Clustering for Visible Targets:* In the proposed approach, the bounding box IDs associated with each visible target are utilized for clustering. To prevent merging between clusters from different visible targets and address issues related to overlapping bounding boxes and coverage discrepancies, relevant criteria are proposed.
3) *Orientation-Aware 2-D-Assignment Merging Algorithm:* A method for calculating the orientation of visible targets based on their bounding boxes and clusters is presented. In addition, a 2-D assignment merging algorithm, which leverages both the class and orientation information of visible targets to mitigate splitting errors, is also proposed.

This article is structured as follows. This work begins with a succinct overview of the existing literature in Section II. Section III formulates the problem statement, describes the measurement models and states the underlying assumptions of this work. Section IV reviews the traditional density-based clustering algorithm, then presents the proposed range-adaptive clustering algorithm and other criteria for radar-only clustering. Section V proposes the camera-assisted clustering method, while Section VI improves the camera-assisted clustering method by addressing additional challenges. Section VII shows the simulation results and analysis of the proposed methods. Section VIII concludes this article and suggests directions for future work.

II. Related Works

Density-based spatial clustering of applications with noise (DBSCAN) is a popular density-based clustering method for radar measurements from extended targets [16]. Such clustering method partitions the data into nested clusters based on a user-specified cluster density threshold. DBSCAN is capable of handling clusters of arbitrary shapes in noisy and large datasets, without requiring any prior knowledge of the number of clusters. Therefore, it is often used as a data-mining technique for real-time applications, such as image reconstruction [17], [18], damage localization [19], and phasor measurement unit (PMU) calibration [20]. However, DBSCAN has a major limitation: it cannot cope with data that have large variations in density, since the cluster density threshold is fixed and specified by the user [21].

To address this issue and improve the performance of DBSCAN, many variants of DBSCAN have been proposed in the literature. For example, Mahran and Mahar [22] developed the GriDBSCAN algorithm, which first clusters the data in multiple axis-parallel grids and then merges them based on shared core points. This algorithm accelerates the clustering process and is robust to different sampling densities. An enhanced version of GriDBSCAN for radar data clustering is presented in [23]. However, this algorithm is not suitable for high-dimensional data. Campello et al. [24] introduced the HDBSCAN algorithm, which extracts the most stable clusters by maximizing the overall clustering stability. Based on this algorithm, Malzer and Baum [25] incorporated cluster-level constraints from background information into HDBSCAN, which improved the clustering results for radar data.

However, the selection of constraints is challenging in dynamic environments. Ram et al. [26] devised the EDBSCAN algorithm, which allows the local density variation within each cluster. Wagner et al. [27] applied the EDBSCAN algorithm to multidimensional radar data and increased the clustering accuracy for pedestrian recognition. Zhang et al. [28] designed the CC-DBSCAN algorithm, which classifies data points with their RGB values and adds class-adaptive constraints to DBSCAN clustering, which outperforms other methods in large-scale point cloud clustering. A comprehensive comparison of other DBSCAN variants is given in [29].

III. Problem Statement

In this work, the main objective is to improve the clustering of the radar measurements using additional camera data. As mentioned in Section I, traditional clustering algorithms often suffer from splitting and merging errors. To mitigate such issues, two following approaches are presented.

1) Radar data are clustered using an adaptive density-based clustering algorithm.
2) Camera data are used as prior information to improve the radar data clustering.

The following assumptions are made in this work.

1) The relative location of the radar and the camera is known.
2) All the intrinsic and extrinsic camera parameters are known precisely.
3) Targets are divided into a few classes with approximately known length and width.
4) Bounding boxes and their classes can be extracted from camera data using an existing object detector.
5) The camera data and radar data are synchronized.

A. System and Measurements Models

Let us denote the state of a scattering point of an extended target at kth time step as $\mathbf{x}_k = [x_k, \dot{x}_k, y_k, \dot{y}_k, z_k, \dot{z}_k]^{\mathrm{T}}$, in which x_k, y_k, and z_k are the 3-D Cartesian coordinates of the scattering point, and $\dot{x}_k$, $\dot{y}_k$, and $\dot{z}_k$ are its velocity components parallel to x-, y-, and z-axes, respectively. Similarly, the state of the radar sensor is denoted by $\mathbf{x}_k^s = [x_k^s, \dot{x}_k^s, y_k^s, \dot{y}_k^s, z_k^s, \dot{z}_k^s]^{\mathrm{T}}$. s and $(\cdot)^{\mathrm{T}}$ denote the radar sensor and the mathematical transpose operation, respectively. The radar sensor provides range, bearing, elevation, and range rate measurements, denoted by

r_k, $\theta_k \in [-\pi, \pi]$, $\gamma_k \in [-(\pi/2), (\pi/2)]$ and $\dot{r}_k$, respectively, and given by

$$r_k = \sqrt{\left(x_k - x_k^s\right)^2 + \left(y_k - y_k^s\right)^2 + \left(z_k - z_k^s\right)^2} \quad (1)$$

$$\theta_k = \tan^{-1}\left(y_k - y_k^s, x_k - x_k^s\right) \quad (2)$$

$$\gamma_k = \tan^{-1}\left(\left(z_k - z_k^s\right), \sqrt{\left(x_k - x_k^s\right)^2 + \left(y_k - y_k^s\right)^2}\right) \quad (3)$$

$$\dot{r}_k = \frac{\begin{array}{c}\left(x_k - x_k^s\right)\left(\dot{x}_k - \dot{x}_k^s\right) + \left(y_k - y_k^s\right)\left(\dot{y}_k - \dot{y}_k^s\right) \\ + \left(z_k - z_k^s\right)\left(\dot{z}_k - \dot{z}_k^s\right)\end{array}}{\sqrt{\left(x_k - x_k^s\right)^2 + \left(y_k - y_k^s\right)^2 + \left(z_k - z_k^s\right)^2}}. \quad (4)$$

Considering $\mathbf{h}(\mathbf{x}_k, \mathbf{x}_k^s) = [r_k, \theta_k, \gamma_k, \dot{r}_k]$, the measurement model can be written as follows:

$$\mathbf{z}_k = \mathbf{h}\left(\mathbf{x}_k, \mathbf{x}_k^s\right) + \mathbf{w}_k \quad (5)$$

where the measurement noise $\mathbf{w}_k$ is the zero-mean Gaussian with measurement covariance $\mathbf{R}_k = \mathrm{diag}(\sigma_{r_k}^2, \sigma_{\theta_k}^2, \sigma_{\gamma_k}^2, \sigma_{\dot{r}_k}^2)$, i.e., $\mathbf{w}_k \sim \mathcal{N}(\mathbf{w}_k; 0, \mathbf{R}_k)$.

Let us denote n_k as the total number of measurements generated by the sensor at time-step k. Using (5), the total number of measurements generated by the sensor at time-step k can be defined as follows:

$$\mathbf{Z}_k = \left\{\mathbf{z}_k^i\right\}_{i=1}^{n_k}. \quad (6)$$

B. Coordinate Transformation

To perform clustering, measurements are often converted to the Cartesian coordinate system from the polar coordinate system using the following equations:

$$x_k = r_k \cos(\theta_k) \cos(\gamma_k) \quad (7)$$

$$y_k = r_k \sin(\theta_k) \cos(\gamma_k) \quad (8)$$

$$z_k = r_k \sin(\gamma_k). \quad (9)$$

In this work, the bias imparted by the polar to Cartesian transformation is considered to be negligible.

IV. Radar-Only Clustering

This section presents the radar measurement clustering approach. The conventional DBSCAN algorithm, based on the works in [16], as well as the required definitions and time complexity, is described in Section IV-A. On the other hand, the proposed modified DBSCAN algorithm with range adaptive criteria is elaborated in Section IV-B.

A. Classical DBSCAN Algorithm

This section introduces a basic version of the DBSCAN algorithm, referred to as the classical DBSCAN Algorithm. For simplicity, the time index k is dropped from the notations of the position and velocity coordinates in this section.

Definition 1: ϵ-neighborhood of a point $l_1 \in P$ is denoted by $N_\epsilon(l_1) = \{l_2 | l_2 \in P, |l_1 - l_2| \leq \epsilon\}$, where P denotes the point cloud.

Definition 2: A point $l_1 \in P$ is a core point if $|N_\epsilon(l_1)| \geq N_{\min}$, where $N_{\min}$ denotes the threshold indicating the minimum number of points required to be present inside the ϵ-neighborhood. Here, $|\cdot|$ denotes the cardinality of the set $N_\epsilon(l_1)$.

Definition 3: A point $l_1 \in P$ is a border point if l_1 exists in the ϵ-neighborhood of a core point but $|N_\epsilon(l_1)| < N_{\min}$.

Definition 4: A point $l_1 \in P$ is a noise point if l_1 is neither a core point nor a border point.

Let us consider two different points $l_1 \in P$ and $l_2 \in P$ with position coordinates (x^{l_1}, y^{l_1}) and (x^{l_2}, y^{l_2}), respectively. l_2 is in the ϵ-neighborhood of l_1 if the Euclidean distance between the x- and y-coordinates of l_1 and l_2 satisfies the following equation:

$$\sqrt{\left(x^{l_1} - x^{l_2}\right)^2 + \left(y^{l_1} - y^{l_2}\right)^2} < \epsilon_{xy} \quad (10)$$

where ϵ_{xy} is the radius threshold.

The time complexity of the classical DBSCAN method is $\Theta(n^2 \cdot D)$, in which n is the number of input radar points, and D is the cost of distance calculation; and the space complexity of DBSCAN is $O(n + I)$, in which I is extra space for index [30].

To negate the splitting and merging clustering errors mentioned in Section I, an adaptive threshold selection is proposed to improve radar-only DBSCAN clustering in Section VI.

B. Range Adaptive Criteria for Radar-Only Clustering

To exploit the radar data effectively, the radar-only clustering takes into account different sampling frequencies on radial and tangential directions, Doppler information, and elevation differences. The variation of radar detection density with range is also considered. The radar-only DBSCAN algorithm is enhanced with the following additional criteria to improve measurement clustering.

1) *Criterion 1:* Considering the Euclidean distance between the x- and y-coordinates of l_1 and l_2, the first neighborhood criterion is defined as follows:

$$\sqrt{\left(x^{l_1} - x^{l_2}\right)^2 + \left(y^{l_1} - y^{l_2}\right)^2} < \epsilon_{xy} \quad (11)$$

where

$$\epsilon_{xy} = \begin{cases} \epsilon_{\mathrm{sr}}, & \text{if } r^{l_1} \leq \alpha_{\mathrm{msr}} \\ \epsilon_{\mathrm{lr}}, & \text{if } r^{l_1} > \alpha_{\mathrm{msr}} \end{cases} \quad (12)$$

where ϵ_{sr} and ϵ_{lr} are the short- and long-range Euclidean distance thresholds, α_{msr} is the maximum value of short range, and r^{l_1} is the range measurement of l_1.

2) Approximate the x- and y-axes direction as radial and tangential directions for different radar points, the following criterion is proposed.

Criterion 2: Considering the x- and y-coordinate differences between the points, the second neighborhood criterion is defined as follows:

$$\left|x^{l_1} - x^{l_2}\right| \leq \epsilon_x \cap \left|y^{l_1} - y^{l_2}\right| \leq \epsilon_y \quad (13)$$

where ϵ_x and ϵ_y are the distance thresholds along the x- and y-axes, respectively. ϵ_x and ϵ_y are defined as follows:

$$\left(\epsilon_x, \epsilon_y\right) = \begin{cases} (\alpha_{\text{srr}}, \alpha_{\text{srt}}), & \text{if } r^{l_1} \leq \alpha_{\text{msr}} \ \& \ \left|x^{l_1}\right| > \left|y^{l_1}\right| \\ (\alpha_{\text{srt}}, \alpha_{\text{srr}}), & \text{if } r^{l_1} \leq \alpha_{\text{msr}} \ \& \ \left|x^{l_1}\right| \leq \left|y^{l_1}\right| \\ (\alpha_{\text{lrr}}, \alpha_{\text{lrt}}), & \text{if } r^{l_1} > \alpha_{\text{msr}} \ \& \ \left|x^{l_1}\right| > \left|y^{l_1}\right| \\ (\alpha_{\text{lrt}}, \alpha_{\text{lrr}}), & \text{if } r^{l_1} > \alpha_{\text{msr}} \ \& \ \left|x^{l_1}\right| \leq \left|y^{l_1}\right| \end{cases} \tag{14}$$

where α_{srr} and α_{lrr} are the short- and long-range radial distance thresholds, and α_{srt} and α_{lrt} are the short- and long-range tangential distance thresholds.

3) The third neighborhood criterion considers the range rate information. l_2 is in the neighborhood of l_1 only in these two cases: 1) both l_1 and l_2 are static points and 2) both l_1 and l_2 are moving points.

To identify whether a radar point is moving or not, the sensor velocity compensated range rate of the point is necessary. Assuming that the x- and y-coordinates of the sensor velocity are $(\dot{x}^s, \dot{y}^s)$, the range rate and bearing measurements of a point $l_1 \in P$ being $\dot{r}^{l_1}$ and θ^{l_1}, the sensor velocity compensated range rate is defined as follows:

$$\dot{R}^{l_1} = \dot{r}^{l_1} + \dot{x}^s \cos\left(\theta^{l_1}\right) + \dot{y}^s \sin\left(\theta^{l_1}\right). \tag{15}$$

As the sensor velocity compensated range rate is known, the third neighborhood criterion is defined as follows.

Criterion 3*:* For the static case, l_2 is in the neighborhood of l_1 if the following equation holds:

$$\left|\dot{R}^{l_1}\right| < \alpha_{\text{rrs}} \cap \left|\dot{R}^{l_2}\right| < \alpha_{\text{rrs}} \tag{16}$$

where α_{rrs} is the sensor velocity compensated range rate threshold to determine whether a point is static or not.

When both l_1 and l_2 are moving points, the following is required to be satisfied:

$$\left|\dot{R}^{l_1}\right| \geq \alpha_{\text{rrs}} \cap \left|\dot{R}^{l_2}\right| \geq \alpha_{\text{rrs}}. \tag{17}$$

In this case, l_2 is considered to be in the neighborhood of l_1 if

$$\Delta\dot{r} = \left|\dot{r}^{l_1} - \dot{r}^{l_2}\right| \leq \epsilon_{\dot{r}} \tag{18}$$

where $\epsilon_{\dot{r}}$ is the range rate difference threshold.

4) *Criterion* 4*:* Considering the elevation difference of l_1 and l_2, l_2 is in the neighborhood of l_1 if

$$\left|z^{l_1} - z^{l_2}\right| \leq \epsilon_z \tag{19}$$

where ϵ_z is the distance threshold along the z-axis.

In addition, $N_{\min}$ is also range adaptive

$$N_{\min} = \begin{cases} N_{\text{sr}}, & \text{if } r^{l_1} \leq \alpha_{\text{msr}} \\ N_{\text{lr}}, & \text{if } r^{l_1} > \alpha_{\text{msr}}. \end{cases} \tag{20}$$

In the proposed method, all of the above criteria are used to conclude whether point l_2 is in the neighborhood of l_1 or not.

V. Simple Camera-Assisted Clustering

This section presents a camera-assisted clustering method that was initially proposed in the conference version of this work [15]. This method is a simple yet effective technique for clustering measurements from closely spaced objects based on their visual features.

In order to map the radar measurements with the associated image information, it is assumed that all the necessary camera parameters are available.

As the first step, objects in the image are required to be identified. A pretrained convolutional neural network model, such as YOLOV5, can be used to obtain both bounding boxes and associated class estimates for all the objects [31]. Once the bounding boxes are found in the image plane, two following options are available: 1) project the bounding boxes to the radar measurement plane and 2) project the radar measurements to the image plane. The latter option is more feasible than the former because the image does not provide depth information. Therefore, the camera-assisted clustering is performed by projecting the radar measurements onto the corresponding image.

A. Projection of Radar Measurements Onto Images

Let us assume that the radar provides n_k number of measurements at time-step k. Considering lth radar measurement, projection onto the image can be performed using the following steps.

1) *Step 1:* Let us denote the Cartesian coordinates of the lth measurement as $[x^l, y^l, z^l]^{\mathrm{T}}$. Sensor coordinate to ego-vehicle (the vehicle on which the sensor is mounted) coordinate conversion is performed as follows:

$$\left[x^{l,e}, y^{l,e}, z^{l,e}, 1\right]^{\mathrm{T}} = \begin{bmatrix} \mathbf{R}^{\text{me}}_{3\times3} & \mathbf{T}^{\text{me}}_{3\times1} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \left[x^l, y^l, z^l, 1\right]^{\mathrm{T}} \tag{21}$$

where e indicates the ego vehicle and me indicates the matrices facilitating the sensor measurement to the ego-vehicle coordinate translation, respectively. Matrices $\mathbf{R}^{\text{me}}_{3\times3}$ and $\mathbf{T}^{\text{me}}_{3\times1}$ denote the rotation and the transformation matrix from the sensor measurement to the ego-vehicle coordinates.

Note that usually $\mathbf{R}^{\text{em}}_{3\times3}$ and $\mathbf{T}^{\text{em}}_{3\times1}$ are assumed to be known a priori, as radar parameters, in which em indicates the matrices facilitating the ego vehicle to the sensor measurement coordinate translation. The relationships between $\mathbf{R}^{\text{me}}_{3\times3}$ and $\mathbf{R}^{\text{em}}_{3\times3}$ and $\mathbf{T}^{\text{me}}_{3\times1}$ and $\mathbf{T}^{\text{em}}_{3\times1}$ are given as follows:

$$\mathbf{R}^{\text{me}}_{3\times3} = \mathbf{R}^{em^T}_{3\times3} \tag{22}$$

$$\mathbf{T}^{\text{me}}_{3\times1} = -\mathbf{T}^{\text{em}}_{3\times1}. \tag{23}$$

2) *Step 2:* In this step, the ego-vehicle coordinates are converted into equivalent camera coordinates using the camera extrinsic matrix. Such conversion in the

homogeneous coordinate system is provided as follows:

$$\left[x^{l,c}, y^{l,c}, z^{l,c}, 1\right]^{\mathrm{T}} = \begin{bmatrix} \mathbf{R}^{\mathrm{ec}}_{3\times3} & \mathbf{T}^{\mathrm{ec}}_{3\times1} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \left[x^{l,e}, y^{l,e}, z^{l,e}, 1\right]^{\mathrm{T}} \tag{24}$$

where c indicates the camera and ec indicates the matrices facilitating the ego-vehicle to the camera coordinate transformation. Note that the whole 4×4 transformation matrix on the right side of (24) is identical to the camera extrinsic matrix, which is assumed to be known as a priori as a camera calibration parameter.

3) *Step 3:* Finally, the camera coordinates are converted into the image pixel coordinates using the camera intrinsic matrix. The transformation is shown as follows:

$$\left[u^l, v^l, 1\right]^{\mathrm{T}} = \frac{1}{z^{l,c}} \begin{bmatrix} \frac{f}{\rho_u} & 0 & u_0 & 0 \\ 0 & \frac{f}{\rho_v} & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \left[x^{l,c}, y^{l,c}, z^{l,c}, 1\right]^{\mathrm{T}} \tag{25}$$

where $[u^l, v^l]^{\mathrm{T}}$ are the projected image pixel coordinates of the lth measurement. f, ρ_u, and ρ_v denote the focal length of the camera and the width and height of a pixel, respectively. u_0 and v_0 represent the coordinates of the image center. Note that the whole 3×4 transformation matrix on the right side of (25) is identical to the intrinsic camera matrix and is assumed to be known as a priori as a camera calibration parameter.

Using the above three steps, all n_k number radar of measurements at time-step k are projected onto the image.

B. Subsequent Camera-Assisted Clustering

The main goal of projecting all n_k radar measurements onto the image frame is to use the additional information (i.e., known bounding boxes and class labels) obtained from the image to improve the clustering.

With target bounding boxes and the projected radar measurements in each image, the radar data can be clustered only based on camera information. However, the lack of depth information from the camera causes severe merging errors in range. To address this drawback, a secondary clustering approach of the projected radar measurements is proposed, which combines both camera and radar data for clustering. In the secondary clustering, the radar measurements are divided by bounding boxes and clustered with range and class adaptive thresholds separately [15].

Still, there are some problems to be solved for this initial camera-assisted algorithm, which are to be discussed in Section V-C.

C. Challenges

The aforementioned algorithm, which integrates camera and radar data, offers a promising improvement in clustering. However, several potential challenges need to be addressed before its practical implementation.

1) Mutually Occluded Objects: In realistic scenarios involving occlusion, some foreground objects may obscure the background objects. Such occlusion results in partial or complete invisibility of the latter in the images. Consequently, a bounding box may encompass not only the projected detections from the visible object but also from the occluded objects. For the latter case, the adaptive parameters chosen based on the class labels of the visible object are inappropriate.

2) Uncertainty in Azimuth and Elevation Measurements: The radar measurements are subjected to uncertainty in azimuth and elevation due to the limited angular resolution and the measurement noise of the radar sensor. This uncertainty causes the projected radar detections in the images to deviate from their true positions, due to the high sensitivity of the camera projection to the angular errors. Consequently, this may lead to incorrect associations between the radar detections and the bounding boxes, such as false outliers or false inliers. Therefore, the effective coverage of the bounding boxes should be enlarged to account for the azimuth and elevation uncertainty of the radar measurements.

3) Overlapping Bounding Boxes: In scenarios involving closely spaced targets, the bounding boxes of two different targets may overlap, especially when the boundaries of the bounding boxes are relaxed to account for the uncertainty in the azimuth and elevation angle measurements. In such cases, the projected detections in the overlapping regions of the bounding boxes are ambiguous, and the camera-only clustering based on the projected detections may produce erroneous results. This may also result in the splitting of the targets. To mitigate such error, the projected detections in the overlapping regions can be assigned to both bounding boxes simultaneously. A secondary clustering is then applied to resolve the resulting ambiguity.

4) Coverage Discrepancy Between Camera and Radar: In some practical applications, the radar sensor may have a larger range coverage than that of the camera sensor. Consequently, not all the radar measurements can be projected onto the images for camera-assisted clustering. In such cases, the radar-only adaptive DBSCAN should be applied to cluster the radar measurements that fall outside the camera's field of view. Conversely, the radar measurements that fall inside the camera's field of view are clustered separately with the camera-assisted clustering. However, this may cause the splitting of clusters from partially visible targets. Therefore, a unified clustering method should be employed for both types of measurements.

VI. Improved Camera-Assisted Clustering

In this section, to address the aforementioned challenges, an improved version of the camera-assisted clustering algorithm is proposed. The pipeline of this algorithm is presented as a flowchart in Fig. 1.

A. Class Adaptive Parameter for Camera-Assisted Clustering

In the proposed camera-assisted adaptive DBSCAN method, *criteria* 1–4 are still applied for ϵ-neighborhood selection to fully utilize radar information. In addition, the target class information are also utilized for clustering.

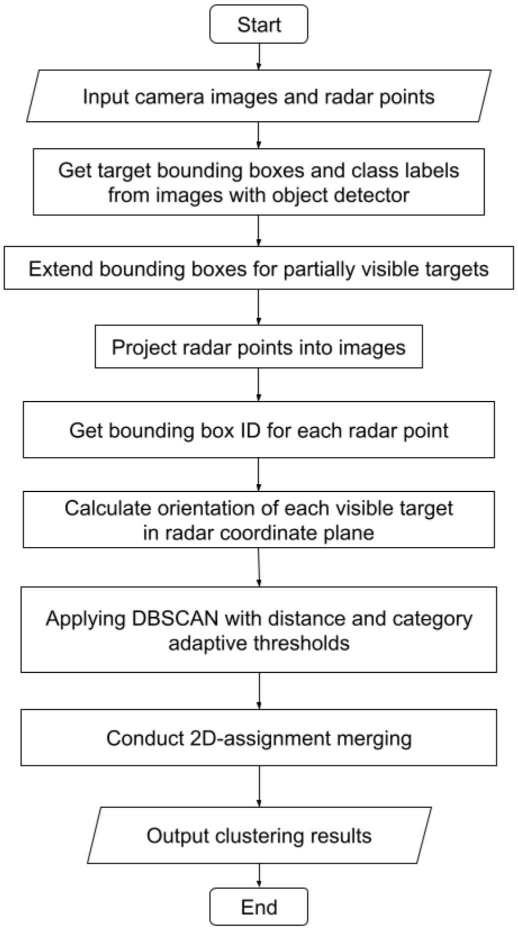

Fig. 1. Flowchart of the proposed video-assisted clustering algorithm.

The miss detection inside a cluster of a large target may lead to the splitting of its clusters. False alarms between clusters of closely spaced small targets can cause the merging of these clusters. To avoid splitting and merging errors in these cases, ϵ_{xy} in (11) and ϵ_x and ϵ_y in (13) changes adaptively to target class. The values of these threshold parameters are set as follows.

1) Bigger for large objects to reduce splitting.
2) Smaller for small objects to reduce merging.

Regarding object occlusion, class information from the camera are precise only for the first target visible to the camera for a given angle. Therefore, only for near points with $r^l \leq \alpha_{\mathrm{msr}}$, the above-mentioned parameters are class adaptive. Relatively, for further points with $r^l > \alpha_{\mathrm{msr}}$, values of these parameters are fixed and different from those for near points.

B. Compensation of Bounding Boxes for Angular Uncertainty

Assuming that elevation measurement error is zero-mean Gaussian distribution with standard deviation σ_γ, and following (7)–(9), the compensated polar coordinates $(r^l, \theta^l, \gamma^l \pm 3\sigma_\gamma)$ can easily be converted into the equivalent Cartesian coordinates $[\bar{x}^l, \bar{y}^l, \bar{z}^l]$. Here, the three-sigma region is considered to cover the possible target region.

From (21)–(25), the projected image pixel coordinates $(\bar{u}^l, \bar{v}^l)$ of $[\bar{x}^l, \bar{y}^l, \bar{z}^l]$ can be calculated. As (u^l, v^l) and $(\bar{u}^l, \bar{v}^l)$ are calculated, the maximum possible measurement offset along the v-axis for measurement l is

$$\delta v^l = \left|\bar{v}^l - v^l\right|. \tag{26}$$

Similarly, assuming that azimuth measurement error is zero-mean Gaussian distribution with standard deviation σ_θ, the compensated polar coordinates $(r^l, \theta^l \pm 3\sigma_\theta, \gamma^l)$ are converted into Cartesian coordinates, and then, they are projected into image pixel coordinates to find the maximum possible measurement offset along the u-axis, δu^l.

The bounding boxes are adjusted by δu^l and δv^l for each measurement to counter possible angular measurement errors.

C. Clustering With Bounding Box ID

Assuming that there are total m bounding boxes in the image with IDs $1, 2, \ldots, m$, two more IDs $m+1$ and $m+2$ are added for the region outside the bounding boxes, but inside the image, and another one for the region outside the camera field-of-view, respectively. Multiple IDs can be associated with each measurement. IDs associated with radar point $l \in P$, id^l, is a set of IDs of all the bounding boxes within which the projected radar point falls, and it is given by

$$\mathrm{id}^l = \begin{cases} \{b_i\}_{i=1}^{M_{\mathrm{ol}}}, & \text{if the projected point is in the overlapping region of bounding boxes } \{b_i\}_{i=1}^{M_{\mathrm{ol}}}\text{, where } M_{\mathrm{ol}} \text{ is the number of overlapping bounding box, and } 1 \leq b_i \leq m \\ \{b_i\}, & \text{if the projected point falls only inside one bounding box } b_i\text{, and } 1 \leq b_i \leq m \\ \{m+1\}, & \text{if the projected point is outside of all bounding boxes} \\ \{m+2\}, & \text{if the projected point is outside of the camera's field-of-view.} \end{cases} \tag{27}$$

Radar points that fall inside the same bounding box can be clustered together.

1) Clustering With Overlapping Bounding Boxes: Considering the possibility of projection points falling in overlapping areas of target bounding boxes, there can be one or more bounding boxes for each measurement.

Radar point l_2 is an ϵ-neighborhood candidate of l_1 if they have a common associated ID, i.e., $\mathrm{id}_{l_1 l_2}^{\mathrm{ints}} = \mathrm{id}^{l_1} \cap \mathrm{id}^{l_2} \neq \emptyset$.

To avoid merging in overlapping areas, class adaptive ϵ in subsequent clustering is chosen as follows:

$$\epsilon = \min\left(\left\{\epsilon_{c_i}\right\}_{i \in \mathrm{id}_{l_1 l_2}^{\mathrm{ints}}}\right) \tag{28}$$

where c_i is the class label of bounding box i and ϵ_{c_i} is the predefined threshold for c_i. In case l_2 satisfies other criteria in subsequent clustering with the above ϵ values, it is an ϵ-neighborhood of l_1.

It is possible for radar points to be assigned to multiple clusters since they have multiple bounding box IDs. To determine the final clusters, the average distances to other points

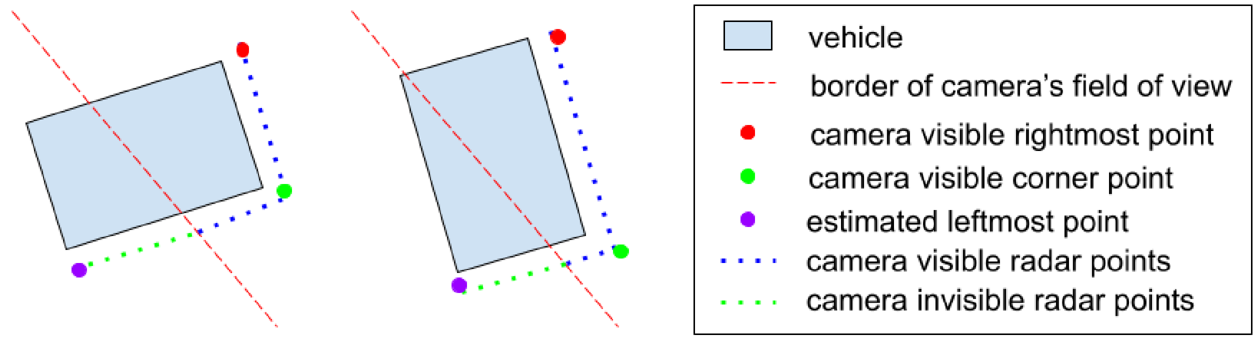

Fig. 2. Different target's visibility scenarios: partially visible target on the left side of the image, with width side fully visible (left) or length side fully visible (right). For a partially visible target on the right side of the image, the scenarios are mirror symmetrical.

in each candidate cluster are calculated for each of these points. Then, these radar points will be attributed to the clusters corresponding to the smallest average distances, i.e., the clusters with the most points that are closest to them.

2) Clustering With Projections Outside Bounding Boxes: If the bounding boxes generated by the object detector do not cover whole objects, some of the radar points might fall outside the bounding box even though they are generated from the same target.

For two radar points l_1 and l_2, if projected l_1 falls in a bounding box but l_2 is outside of all bounding boxes, still l_2 is a possible ϵ-neighborhood candidate of l_1 depending on their distance. In this situation, class adaptive ϵ in subsequent clustering is chosen as follows:

$$\epsilon = w_{\text{out}} \min\left(\{\epsilon_{c_i}\}_{i\in \text{id}^{l_1}}\right) \tag{29}$$

where $w_{\text{out}} \leq 1$ is a decay weight for radar points corresponding to outliers since they are less likely from the target. w_{out} is set as 0.5 in implementation.

With this criterion, splitting caused by inaccurate bounding boxes could be reduced.

D. Extended Bounding Boxes for Partially Visible Targets

To prevent the merging of clusters, points inside bounding boxes are excluded from clusters that have a core point outside the camera's field of view. However, this may result in the fragmentation of clusters belonging to partially visible objects. Therefore, the bounding boxes of such objects are enlarged to mitigate the issue of coverage inconsistency.

We consider partially visible objects that appear on either side of the image. For the left side, we assume that a bounding box with id i $(1 \leq i \leq n)$ belongs to a partially visible object if its left border is less than $a_{\min}$ pixels away from the left border of the image. This is considered as the left boundary of the camera's field of view. The parameter $a_{\min}$ can be set to a positive value to account for possible errors in the bounding box.

The extension of a box requires the estimation of the object's distance (or range) from the camera. This is done by using the radar measurements that are contained within the bounding box. Assuming that the right border of the bounding box is fully visible in the image, one or more radar measurements that are close to or on the right border of the bounding box are chosen. Any interference points with a very high range estimate are discarded. Denoting the average coordinates of the selected points (along the x- and y-axes) by $\bar{x}^r$ and $\bar{y}^r$, The rightmost bottom corner point of the target, denoted by l^{br}, is then computed as follows:

$$l^{\text{br}} = \left(\bar{x}^r, \bar{y}^r, 0\right). \tag{30}$$

Similarly, the coordinates of the corner that is closest to the ego vehicle can be found, and it is given by

$$l^c = \left(\bar{x}^c, \bar{y}^c, 0\right). \tag{31}$$

As the class c_i is known from the camera, there are two different situations for the target's visibility, as shown in Fig. 2.

1) The width side of the target is fully visible in the image, i.e., $D(l^{\text{br}}, l^c) \leq w_{c_i} + b$.
2) The length side of the target is fully visible in the image, i.e., $D(l^{\text{br}}, l^c) > w_{c_i} + b$.

where $D(l^{\text{br}}, l^c)$ is the Euclidean distance between point l^{br} and l^c, w_{c_i} is the width of class c_i, and b is a preset parameter to factor the errors in l^{br} and l^c.

Define the target's orientation angle $\theta_i \in (-(\pi/2), (\pi/2)]$ as the angle between the length direction of the target and the y-axis positive direction of the ego coordinate, measured counterclockwise. Depending on different target visibilities, it can be expressed as follows:

$$\theta_i^l = \begin{cases} \tan^{-1}\left(\bar{y}^r - \bar{y}^c, \bar{x}^r - \bar{x}^c\right), & D\left(l^{\text{br}}, l^c\right) \leq w_{c_i} + b \\ \tan^{-1}\left(\bar{x}^r - \bar{x}^c, \bar{y}^r - \bar{y}^c\right), & D\left(l^{\text{br}}, l^c\right) > w_{c_i} + b. \end{cases} \tag{32}$$

Then, the Cartesian coordinate of the target's bottom left corner point l^{bl} can be expressed as follows:

$$x^{\text{bl}} = \begin{cases} \bar{x}^r - l_{c_i}\sin\theta_i^l - w_{c_i}\cos\theta_i^l, & D\left(l^{\text{br}}, l^c\right) \leq w_{c_i} + b \\ \bar{x}^r + l_{c_i}\sin\theta_i^l - w_{c_i}\cos\theta_i^l, & D\left(l^{\text{br}}, l^c\right) > w_{c_i} + b \end{cases} \tag{33}$$

$$y^{\text{bl}} = \begin{cases} \bar{y}^r + l_{c_i}\cos\theta_i^l - w_{c_i}\sin\theta_i^l, & D\left(l^{\text{br}}, l^c\right) \leq w_{c_i} + b \\ \bar{y}^r - l_{c_i}\cos\theta_i^l - w_{c_i}\sin\theta_i^l, & D\left(l^{\text{br}}, l^c\right) > w_{c_i} + b \end{cases} \tag{34}$$

$$z^{b1} = 0 \tag{35}$$

where l_{c_i} is the length of class c_i.

From (21)–(25), the projected image pixel coordinates $(u^{\text{bl}}, v^{\text{bl}})$ of l^{bl} can be calculated; then, u^{bl} is the new minimum u coordinate of the extended bounding box, also the new lower u boundary of image projection range. Note that the value of this u is less than 0.

Similarly, if the right border of the bounding box i is within $a_{\min}$ pixels of the right border of the image, different target's visibility situations are just the mirror symmetry of Fig. 2. The Cartesian coordinate of the target's bottom right corner point l^{br} of the extended box can be expressed as follows:

$$x^{\text{br}} = \begin{cases} \bar{x}^l + l_{c_i}\sin\theta_i^r + w_{c_i}\cos\theta_i^r, & D\left(l^{\text{bl}}, l^c\right) \le w_{c_i} + b \\ \bar{x}^l - l_{c_i}\sin\theta_i^r + w_{c_i}\cos\theta_i^r, & D\left(l^{\text{bl}}, l^c\right) > w_{c_i} + b \end{cases} \tag{36}$$

$$y^{\text{br}} = \begin{cases} \bar{y}^l - l_{c_i}\cos\theta_i^r + w_{c_i}\sin\theta_i^r, & D\left(l^{\text{bl}}, l^c\right) \le w_{c_i} + b \\ \bar{y}^l + l_{c_i}\cos\theta_i^r + w_{c_i}\sin\theta_i^r, & D\left(l^{\text{bl}}, l^c\right) > w_{c_i} + b \end{cases} \tag{37}$$

$$z^{\text{br}} = 0 \tag{38}$$

where l^{bl} is the bottom left corner point, and

$$\theta_i^r = \begin{cases} \tan^{-1}\left(\bar{y}^c - \bar{y}^l, \bar{x}^c - \bar{x}^l\right), & D\left(l^{\text{bl}}, l^c\right) \le w_{c_i} + b \\ \tan^{-1}\left(\bar{x}^c - \bar{x}^l, \bar{y}^c - \bar{y}^l\right), & D\left(l^{\text{bl}}, l^c\right) > w_{c_i} + b. \end{cases} \tag{39}$$

The projected image pixel coordinates $(u^{\text{br}}, v^{\text{br}})$ of l^{br} can be calculated; then, u^{br} is the new maximum u-coordinate of the extended bounding box.

If the points in the cluster are sparse or the orientation angle of targets is close to 0 or $(\pi/2)$, the calculated leftmost or rightmost point may coincide with the calculated corner point. In this case, the bounding box is extended to the maximum possible length, and the bottom left corner point l^{bl} is given by

$$x^{\text{bl}} = x^{\text{br}} - l_{c_i} \tag{40}$$
$$y^{\text{bl}} = y^{\text{br}}. \tag{41}$$

The bottom right corner point l^{br} is given by

$$x^{\text{br}} = x^{\text{bl}} + l_{c_i} \tag{42}$$
$$y^{\text{br}} = y^{\text{bl}}. \tag{43}$$

With this proposed method, radar points out of the camera's field-of-view but inside the extended bounding box can be clustered with other points from the same object, while the other invisible points falling outside the extended boxes are clustered as in Section VI-C2.

E. Calculation of Object Orientation

Knowing the orientation of the targets adds additional information while performing the merging, as described in Section VI-F. In this section, estimating the orientation using the camera data is explained. With the bounding box and corresponding class information, the leftmost, rightmost, and corner points of the corresponding cluster can be found, as mentioned in Section VI-D. The orientation angle θ_i of the visible target can be calculated as follows.

In the case of partially visible bounding boxes, (32) and (39) can be used to find the orientation. There is one more case in which both sides of the target are partially visible, i.e., $D(l^{\text{br}}, l^c) < w_{c_i} - b$ or $D(l^{\text{bl}}, l^c) < w_{c_i} - b$. Hence, the width and length cannot be identified from them. Although this case is merged into one of the previous cases when considering the extension of bounding boxes, θ_i is actually ambiguous in this case. If $\phi = \theta_i^r$, or θ_i^l, then the orientation angle is given by

$$\theta_i = \begin{cases} \phi \text{ or } \phi - \dfrac{\pi}{2}, & \phi \ge 0 \\ \phi \text{ or } \phi + \dfrac{\pi}{2}, & \phi < 0. \end{cases} \tag{44}$$

In the case of the target fully inside the camera field-of-view, it is possible that only one side of the target is visible to the radar while the other sides are totally blocked out. In this situation, the "corner point" calculated as in Section VI-D is actually on the only visible side, and the length of the visible side is the Euclidean distance between the leftmost and rightmost points $D(l^{\text{bl}}, l^{\text{br}})$.

The way to distinguish this case from the case when both sides of the target are visible with the corresponding cluster is to calculate the perpendicular distance from the calculated "corner point" to the connection line of the leftmost and rightmost points.

The orientation angle θ_{lr} of the connection line of the leftmost and rightmost point is given by

$$\theta_{\text{lr}} = \tan^{-1}\left(y^{\text{br}} - y^{b1}, x^{\text{br}} - x^{b1}\right). \tag{45}$$

The perpendicular distance from the calculated "corner point" to the connection line is given by

$$l_{\text{perp}} = \left|\left(-x^c\sin\theta_{\text{lr}} + y^c\cos\theta_{\text{lr}}\right) - \left(-x^{\text{br}}\sin\theta_{\text{lr}} + y^{\text{br}}\cos\theta_{\text{lr}}\right)\right|. \tag{46}$$

If $l_{\text{perp}} \le d_{\text{perp}}$, where d_{perp} is a distance threshold to distinguish these two cases, then the calculated "corner point" is considered as on the connection line and actually only one side of the target is visible. In this situation, δx and δy are given by

$$\delta x = x^{\text{br}} - x^{\text{bl}} \tag{47}$$
$$\delta y = y^{\text{br}} - y^{\text{bl}}. \tag{48}$$

If $l_{\text{perp}} > d_{\text{perp}}$, then the calculated "corner point" is a real corner point of the target, and both sides of the target are visible. In this case, define

$$l_{\max} = \max\left(D\left(l^{\text{bl}}, l^c\right), D\left(l^c, l^{\text{br}}\right)\right) \tag{49}$$
$$l_{\min} = \min\left(D\left(l^{\text{bl}}, l^c\right), D\left(l^c, l^{\text{br}}\right)\right) \tag{50}$$

and δx and δy is given by

$$\delta x = \begin{cases} \bar{x}^c - \bar{x}^l, & \text{if } D\left(l^{\text{bl}}, l^c\right) \ge D\left(l^c, l^{\text{br}}\right) \\ \bar{x}^r - \bar{x}^c, & \text{if } D\left(l^{\text{bl}}, l^c\right) < D\left(l^c, l^{\text{br}}\right) \end{cases} \tag{51}$$

$$\delta y = \begin{cases} \bar{y}^c - \bar{y}^l, & \text{if } D\left(l^{\text{bl}}, l^c\right) \ge D\left(l^c, l^{\text{br}}\right) \\ \bar{y}^r - \bar{y}^c, & \text{if } D\left(l^{\text{bl}}, l^c\right) < D\left(l^c, l^{\text{br}}\right). \end{cases} \tag{52}$$

Then, the orientation θ_i is calculated as follows:

$$\theta_i = \begin{cases} \tan^{-1}(-\delta x, \delta y), & \text{if } l_{\max} > w_{c_i} + b \\ \tan^{-1}(\delta y, \delta x), & \text{if } w_{c_i} - b \le l_{\max} \le w_{c_i} + b \\ & \text{and } D\left(l^{\mathrm{br}}, l^{c}\right) < w_{c_i} - b. \end{cases} \tag{53}$$

When $l_{\max}$ is not long enough to decide whether that is length or width, i.e., $w_{c_i} - b \le l_{\min} \le l_{\max} \le w_{c_i} + b$ or $l_{\max} \le w_{c_i} - b$, (44) can be used to resolve the ambiguity.

F. 2-D Assignment-Based Merging Algorithm

Sometimes, accurate clustering cannot be achieved without an additional merging or splitting step. The proposed criteria (28) and (29) reduce merging but may introduce splitting. Hence, a merging algorithm is proposed to identify the split clusters and merge them.

The merging problem can be formulated into a 2-D assignment problem. Given N clusters that can be merged with other clusters or remain unchanged, the objective is to find an optimal assignment to minimize the cost function, which is given by

$$\min \sum_{i=0}^{N} \sum_{j=0}^{N} c_{i,j} b_{i,j} \tag{54}$$

$$\text{s.t.:} \sum_{j=1}^{N} b_{i,j} = 1 \quad \forall\ 1 \le i \le N \tag{55}$$

$$\sum_{i=1}^{N} b_{i,j} = 1 \quad \forall\ 1 \le j \le N \tag{56}$$

$$b_{i,j} \in \{0, 1\} \quad \forall\ b_{i,j}. \tag{57}$$

Here, $b_{i,j} = 1$ indicates cluster i and j should be merged; otherwise, they are from different targets and should not be merged. $b_{i,0}$ or $b_{0,i}$ indicates that the cluster i is not merged with any other cluster. The cost $c_{i,j}$ is the cost of merging clusters i and j and is calculated as described below. The calculated orientation and the class obtained from the camera data are used to decide if radar clusters should be merged or not.

1) *Cost of Assigning Cluster $i(>0)$ to $j(>0)$:* If $D_{\min}(i, j) > \theta_{d\min}$, $D_{\max}(i, j) > \theta_{d\max}$ or one cluster is inside and the other outside of the camera field-of-view

$$c_{i,j} = \infty. \tag{58}$$

Otherwise,

$$c_{i,j} = D_{\min}(i, j) + \left|D_{\max}(i, j) - \mathrm{diag}_c\right| + \left|l_{\mathrm{proj}} - l_c\right| + \left|w_{\mathrm{proj}} - w_c\right|. \tag{59}$$

In the above

$$D_{\min}(i, j) = \min(D(l_1, l_2)) \quad \forall l_1 \in C_i\ \forall l_2 \in C_i,\ l_1 \ne l_2 \tag{60}$$

$$D_{\max}(i, j) = \max(D(l_1, l_2)) \quad \forall l_1 \in C_i\ \forall l_2 \in C_i,\ l_1 \ne l_2 \tag{61}$$

where $D(l_1, l_2)$ is the Euclidean distance between points l_1 and l_2, c is the class of the target that C_i associated with and l_c and w_c are the length and width of class c, respectively, and l_{proj} and w_{proj} are the projected length and width of the targets in the orientation direction.

$D_{\min}(i, j)$ is added to the cost to encourage merging closer clusters. The value of $D_{\max}(i, j)$ should be close to the length of the target's diagonal, which is given by

$$\mathrm{diag}_c = \sqrt{{l_c}^2 + {w_c}^2}. \tag{62}$$

If C_i is not associated with an image bounding box, then the class is not known. In this case, the parameters of the most common target type, e.g., car, can be used. $|D_{\max}(i, j) - \mathrm{diag}_c|$ is included in the cost to minimize the difference between the expected diagonal distance and the observed distance.

The projected length of the cluster after merging is

$$\begin{aligned} l_{\mathrm{proj}} = {} & \max\left(x^{l_1}\cos\psi + y^{l_1}\sin\psi, x^{l_2}\cos\psi + y^{l_2}\sin\psi\right) \\ & - \min\left(x^{l_1}\cos\psi + y^{l_1}\sin\psi, x^{l_2}\cos\psi + y^{l_2}\sin\psi\right) \\ & \forall l_1 \in C_i \quad \forall l_2 \in C_i \end{aligned} \tag{63}$$

where ψ is the target orientation measured from x axis. If C_i is a visible cluster with orientation angle θ_i, then, ψ is given by

$$\psi = \begin{cases} \theta_i - \pi, & \text{if } \theta_i \ge 0 \\ \theta_i + \dfrac{\pi}{2}, & \text{if } \theta_i < 0. \end{cases} \tag{64}$$

Otherwise, $\psi \in (-(\pi/2), (\pi/2)]$ is the angle of line connecting the centers of C_i and C_j. Similarly, the projected width of the cluster after merging is

$$\begin{aligned} w_{\mathrm{proj}} & = \max\left(-x^{l_1}\sin\psi + y^{l_1}\cos\psi, -x^{l_2}\sin\psi + y^{l_2}\cos\psi\right) \\ & - \min\left(-x^{l_1}\sin\psi + y^{l_1}\cos\psi, -x^{l_2}\sin\psi + y^{l_2}\cos\psi\right) \\ & \forall l_1 \in C_i\ \forall l_2 \in C_i. \end{aligned} \tag{65}$$

If the orientation angle is ambiguous, take the value that minimizes the cost for $c_{i,j}$, and fix the orientation angle as this for the new cluster if C_i and C_j are merged.

2) *Cost of Dummy:* In this case, the cluster remains unchanged. Therefore,

$$c_{i,0} = c_{0,i} = c_d \quad \forall\ 1 \le i \le N \tag{66}$$

where c_d, which is selected based on tolerance for each term in (60), is the cost for any cluster to remain constant.

3) *Self-Assignment Cost:* To avoid associating a cluster to itself, the cost of $c_{i,i}$ is set to ∞ for all i.

Then, this 2-D assignment problem can be solved with existing methods, such as the auction algorithm.

The proposed merging algorithm will be iterated several times until all the clusters remain unchanged.

In Section VI-G, the time and space complexity analysis of the proposed camera-assisted clustering algorithm is introduced.

G. Time Complexity Analysis of the Proposed Clustering Algorithm

Let n and m be the number of radar measurements and the number of detected targets in the image frame, respectively. Both the methods for projection of the radar measurements onto the images and angular uncertainty compensation, discussed in detail in Sections V and VI-B, require a runtime complexity of $\Theta(n)$. Runtime complexity for clustering while using bounding box ID, introduced in Section VI-C, is $\Theta(n^2)$. Note that the distance evaluation to determine whether the projected point falls inside the cluster is independent of m. Extending the bounding boxes for partially visible targets and the orientation evaluations, introduced in Section VI-D, has the complexity of $\Theta(m)$ each. Time complexity associated with the 2-D assignment-based merging, from Section VI-F, can be noted as $O(c^3 \log c)$, where c is the number of clusters before merging. Note that the runtime complexity of the auction algorithm is $O(c^2 \log c)$ and the merging algorithm can iterate for at most c times [32].

Evidently, the number of clusters c cannot be larger than the number of radar measurements n. Moreover, the radar measurements that are projected into different bounding boxes are divided into various clusters before merging. Therefore, $m \leq c \leq n$. Thus, the total runtime complexity for the proposed algorithm is $O(n^3 \log n)$.

VII. Results

This section details the performance evaluations of the proposed algorithms, utilizing simulated data alongside two publicly available autonomous driving datasets, namely, nuScenes [33] and Boreas [34]. Comprehensive documentation of the data acquisition protocols, as well as the camera parameters including the intrinsic and extrinsic matrices for both datasets, are available in [33] and [34]. Measurements generated from moving targets, e.g., moving vehicles and pedestrians, are considered in this work.

A. Experimental Settings and Datasets

To evaluate the performance of the proposed clustering methods, four following cases are considered.

1) *Radar-Only Clustering Without Proposed Parameters:* Radar measurements are clustered using DBSCAN with only range-adaptive ϵ and $N_{\min}$, as discussed in Section IV-A.
2) *Radar-Only Clustering:* Radar measurements are clustered using the range adaptive DBSCAN method with proposed parameters (see Section IV-A for detailed discussion).
3) *Camera-Assisted Clustering Without Merging Step:* Following the detailed discussions from Section V, radar measurements are projected onto the images to perform secondary clustering with extra information from the camera.
4) *Camera-Assisted Clustering With Merging Step:* To evaluate the effect of our proposed merging algorithm, camera-assisted clustering without and with the merging step is considered separately.

TABLE I
SIMULATED RADAR PARAMETERS

Parameter	Value
Range Resolution	1.0 m
Range Rate Resolution	2.0 m/s
Azimuth Resolution	3.0 deg
Detection Probability	0.8
False Alarm Rate	1×10^{-4}

TABLE II
PARAMETERS OF RADAR-ONLY CLUSTERING

Parameter	Value for simulation	Value for dataset
ϵ_{sr}	1.5 m	4.0 m
N_{sr}	2	2
ϵ_{lr}	5.0 m	5.0 m
N_{lr}	1	2
α_{msr}	20.0 m	40.0 m
α_{srr}	1.0 m	2.0 m
α_{srl}	1.5 m	4.0 m
α_{lrr}	2.0 m	3.0 m
α_{lrl}	4.0 m	5.0 m
$\epsilon_{\dot{r}}$	5.0 m	5.0 m
ϵ_z	3.0 m	3.0 m

The following metrics are used for evaluation.

1) *Percentage of Correct Clustering:* This is considered as the percentage of the correctly clustered points in all radar measurements. The ground truth is known in MATLAB simulation. For the real datasets, clusters are labeled manually with the help of image data, as the ground truth is unavailable.
2) *Average Number of Splitting per Frame:* If a cluster is split into m clusters, then this is recorded as $m-1$ times of splitting.
3) *Average Number of Merging per Frame:* If m clusters are merged into one cluster, then this is recorded as $m-1$ times of merging. If false alarms are merged into nearby clusters, then a time of merging is recorded for each false alarm.

B. MATLAB Simulation Results

To evaluate the performance of our proposed method with MATLAB simulation, 40 frames of data from a simulated scenario in which the ego car drives straight in dense traffic is used. The parameters of the simulated 3-D Doppler radar are shown in Table I. The selected parameters for radar-only clustering are shown in Table II.

Fig. 3 shows the clustering results of the radar-only algorithm with and without the proposed parameters, and Table III shows the numerical evaluation of these results. Note that the projection of the radar measurements onto the image corresponding to radar-only clustering, shown in Fig. 3(a) and (b), is provided for analysis only. No additional information obtained by the camera is used for clustering, and this is kept for other such figures. The proposed radar-only clustering approach significantly reduces the merging error but slightly increases the splitting error.

Even with the proposed parameters, there is a trade-off between the number of splits and merges. The clustering

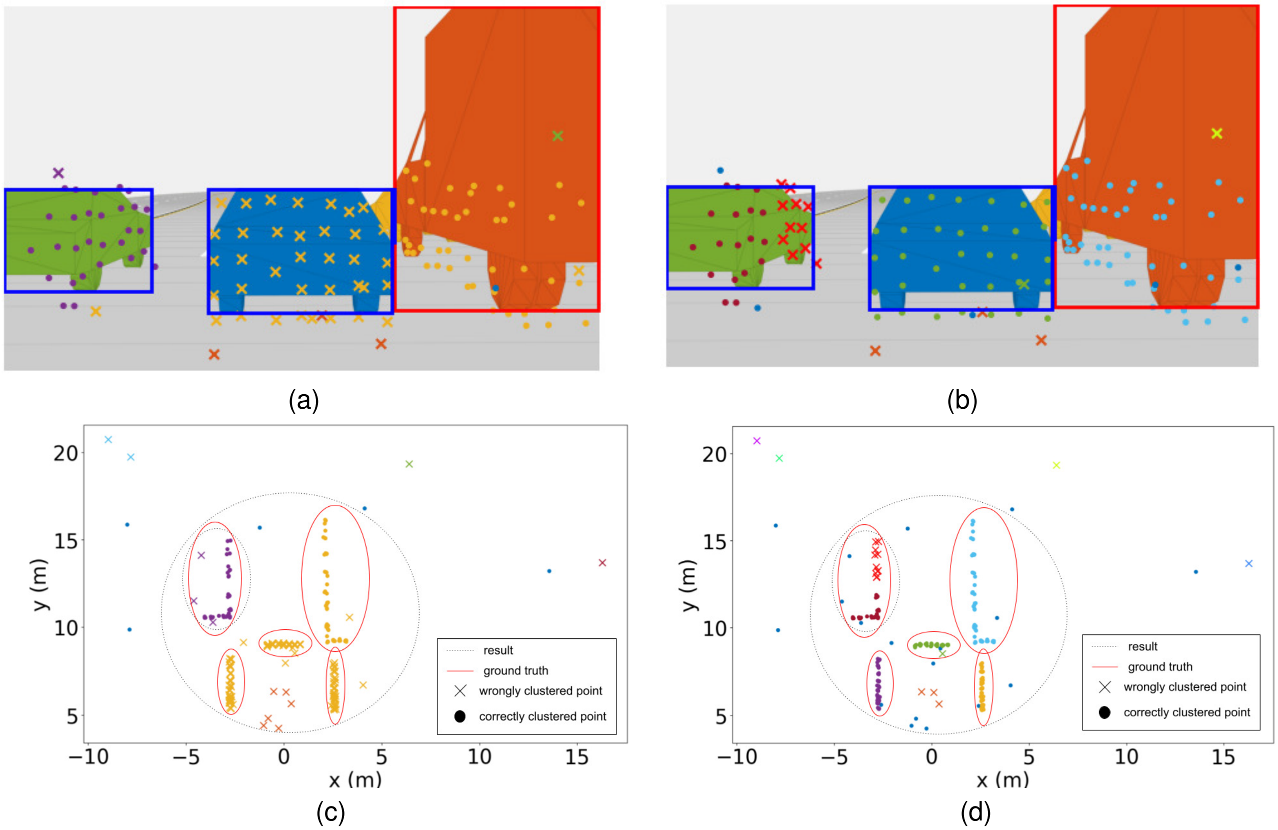

Fig. 3. Clustering results on simulated data: projection of radar measurements onto the images and associated clustering. (a) and (c) Radar-only clustering without proposed parameters. (b) and (d) Radar-only clustering with the proposed parameters. Note that different colors in (c) and (d) represent different cluster IDs in the results, which is kept for other such figures.

TABLE III

PERFORMANCE EVALUATIONS OF RADAR-ONLY CLUSTERING ON SIMULATED DATA

Metric	**Radar-only clustering without proposed parameters**	**Radar-only clustering with proposed parameters**
Correct clustering (%)	80.00	92.11
Average number of splitting	0.0	1.53
Average number of merging	6.90	1.05

TABLE IV

PARAMETER VALUES FOR REDUCING MERGING OR SPLITTING

Parameter	P1	P2
$\epsilon_{sr}(m)$	3.0	3.0
$\alpha_{srr}(m)$	0.5	2.0
$\alpha_{srl}(m)$	1.0	3.0
$\alpha_{lrr}(m)$	1.0	3.0
$\alpha_{lrl}(m)$	2.0	5.0

results with the parameters provided in Table IV are shown in Fig. 4 and Table V. The smaller parameter values of parameter set $P1$ cause splitting, and the larger parameter values of parameter set $P2$ introduce merging.

Next, the performance of the proposed camera-assisted clustering algorithm is tested with the simulated data. Fig. 5 shows the result of clustering using the aforementioned radar-only and camera-assisted methods. The values of class adaptive parameters are shown in Table VI, and the sizes of different types of targets are shown in Table VII.

As shown in Fig. 5(a) and (c), the radar-only algorithm fails to split the clusters from the blue car and the orange truck close to each other since the epsilon values are too large to splitting them. However, decreasing the epsilon values in the radar-only algorithm may lead to the splitting of the further parts of clusters from the green car and the orange truck in the result. Meanwhile, some noise points are wrongly clustered. In comparison, as shown in Fig. 5(b) and (d), with bounding box information from the image, the camera-assisted method splits all clusters accurately and reduces clustering of noise points. The extended bounding boxes frame the largest area including detections from the corresponding target in the image; thus, they reduce splitting for clusters of partially visible targets. Note that the radar-only algorithm is also able to cluster the clusters of partially visible targets accurately since the detections from the invisible parts are dense and easy to cluster. The necessity of extending the bounding box is shown further in the experiments on the dataset.

Fig. 6 shows the significance of the merging step. With the same parameters for adaptive DBSCAN, sometimes splitting

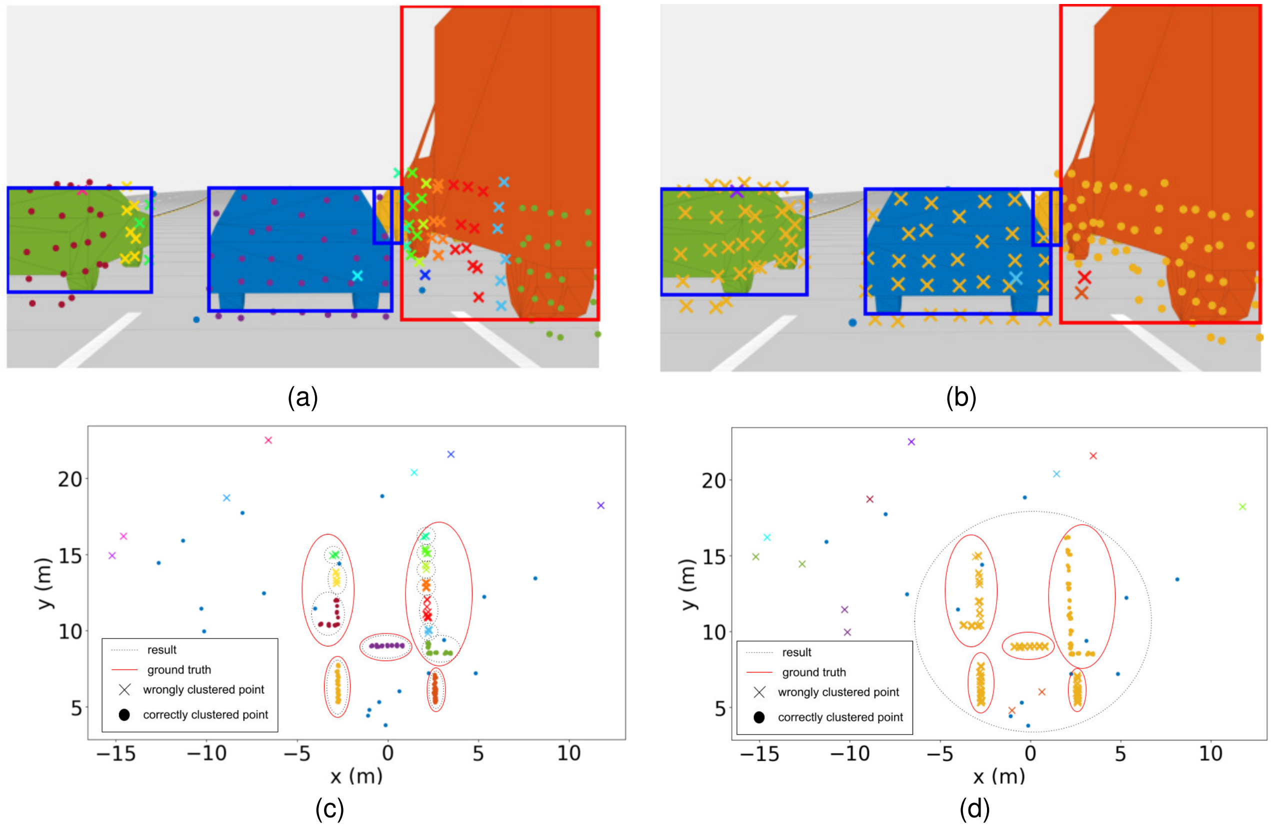

Fig. 4. Clustering results on simulated data: projection of radar measurements onto the images and associated clustering. (a) and (c) Radar-only clustering with parameter set $P1$. (b) and (d) Radar-only clustering with parameter set $P2$.

TABLE V
PERFORMANCE EVALUATIONS FOR RADAR-ONLY CLUSTERING OF SIMULATION DATA WITH DIFFERENT PARAMETER SETTINGS

Metric	Radar-only clustering with P1 parameters	Radar-only clustering with P2 parameters
Correct clustering (%)	78.00	69.00
Average number of splitting	5.15	0.05
Average number of merging	0.40	2.25

TABLE VI
TARGET CLASS-SPECIFIC PARAMETERS IN SIMULATION

Target class	$\epsilon_{sr}(m)$	$N_{min_{sr}}$	$\alpha_{srr}(m)$	$\alpha_{srl}(m)$
Car	1.5	2	1.0	1.5
Truck	2.0	2	1.0	2.0
Bicycle	1.0	1	0.3	0.5
Pedestrian	1.0	1	0.3	0.5

TABLE VII
SIZES OF TARGET CLASSES

Target class	Length (m)	Width (m)
Car	4.7	1.8
Bus	8.0	3.2
Truck	8.1	2.45
Bicycle	1.0	1.0
Motorcycle	2.3	0.9
Pedestrian	0.25	0.4

still occurs in the clustering results of the camera-assisted algorithm because of miss detection, as shown in Fig. 6(a) and (c). To merge these split clusters precisely, an additional merging step, as described in Section VI-F, is added. The clustering result with merging is shown in Fig. 6(c) and (d), in which the proposed merging step merges all the split clusters without introducing more merging errors to other clusters.

To numerically evaluate the simulation performance of the proposed clustering approach, the results are presented using the aforementioned metrics for all of the three clustering approaches in Table VIII.

As shown in Table VIII, with bounding box information and extended bounding box introduced in clustering, the clustering accuracy is improved, and the merging error is sharply reduced. However, these clustering steps reduce the splitting error only slightly. The additional merging step further increases the clustering accuracy and significantly reduces the splitting error without bringing up the merging error. Since in the simulation scenarios, vehicles are all within 30 m, and the resolution of radar is relatively high, the results of our proposed method are not significantly better than those of the radar-only algorithm.

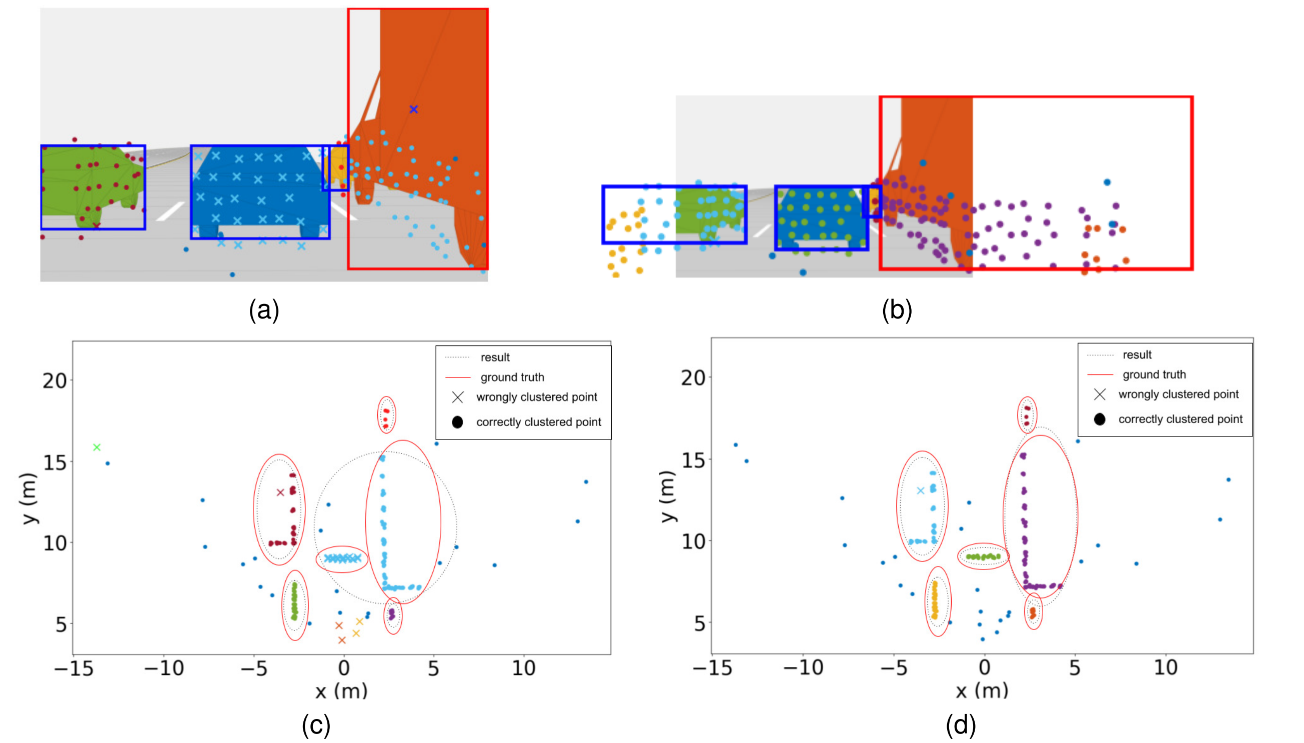

Fig. 5. Clustering results on simulated data: projection of radar measurements onto the images and associated clustering. (a) and (c) Radar-only clustering. (b) and (d) Camera-assisted clustering without merging.

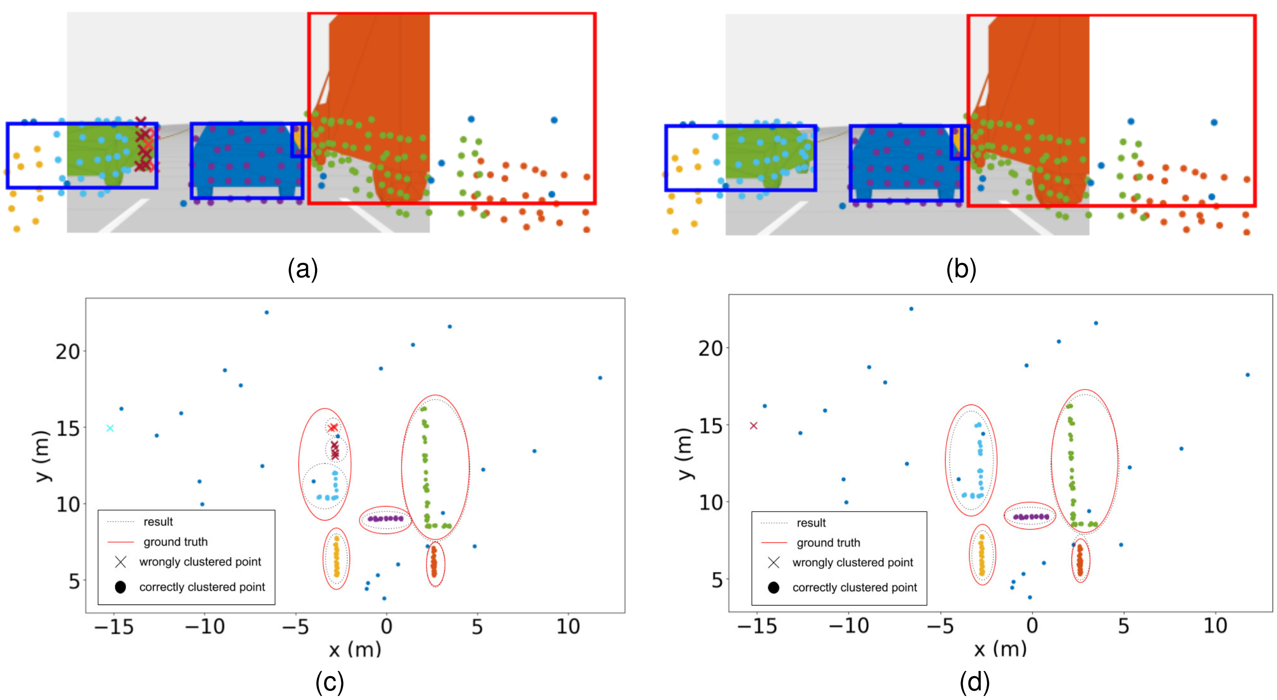

Fig. 6. Clustering results in simulation: projection of radar measurements onto the images and associated clustering. (a) and (c) Camera-assisted clustering without merging. (b) and (d) Camera-assisted clustering with merging.

TABLE VIII
PERFORMANCE EVALUATIONS ON SIMULATION DATA

Metrics	**Radar-only Clustering**	**Camera-assisted without merging**	**Camera-assisted with merging**
Correct clustering (%)	92.11	93.69	96.78
Average number of splitting	1.525	1.3	0.525
Average number of merging	1.05	0.675	0.675

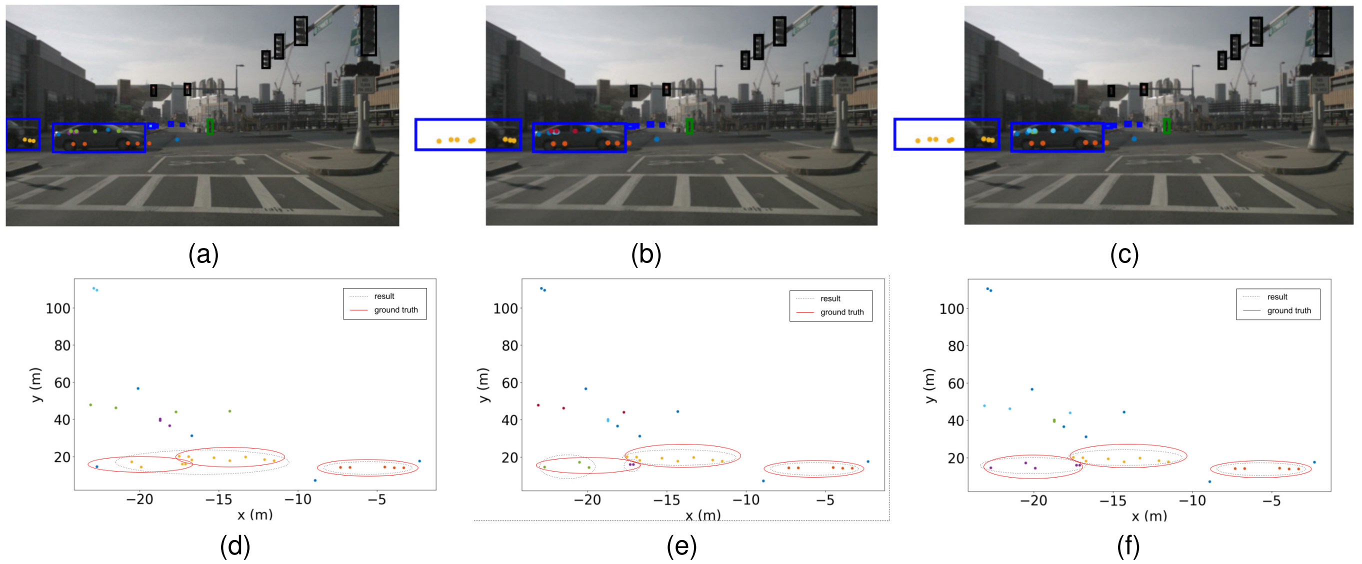

Fig. 7. Clustering results on the nuScenes dataset: projection of radar measurements onto the images and associated clustering. (a) and (d) Radar-only clustering. (b) and (e) Camera-assisted clustering without merging step. (c) and (f) Camera-assisted clustering with merging step.

TABLE IX
TARGET CLASS-SPECIFIC PARAMETERS FOR REAL DATA

Target class	$\epsilon_{sr}(m)$	$N_{min_{sr}}$	$\alpha_{srr}(m)$	$\alpha_{srl}(m)$
Pedestrian	1.0	1	0.5	0.5
Bicycle	2.0	1	1.0	2.0
Car	4.0	1	2.5	4.0
Motorcycle	2.0	1	1.0	2.0
Bus	6.0	2	2.5	5.0
Truck	5.0	2	2.5	4.5

C. Experimental Results on Real Datasets

1) Experimental Results on nuScenes Dataset: To evaluate the performance of our proposed method in real scenarios, 22 frames of data with more partially visible targets and manually identifiable radar detection clusters are selected from scene-0553 and scene-0757 in nuScenes dataset and used in our experiments.

Since the radar parameters are different for the real dataset, another set of parameters for clustering is set up. The parameters in radar-only clustering on the real dataset are shown in Table II. The class adaptive parameters in camera-assisted clustering are shown in Table IX. The sizes of targets are the same as that shown in Table VII.

Fig. 7 shows the result of clustering radar measurements on the dataset using the aforementioned radar-only and camera-assisted methods.

In nuScenes dataset, vehicles are further away (out of 20 m), and the azimuth resolution of radar is lower than in MATLAB simulation scenarios. Sometimes, the distance between clusters of two objects that are close to each other is nearly the same as the distance between two points from the same cluster. To cluster radar measurements in low resolution and further distance, ϵ values are set larger in adaptive DBSCAN, which leads to more merging errors in the clustering result of the radar-only algorithm. For example, as shown in Fig. 7(a) and (d), the cluster of the partially visible car on the left is merged with another cluster from an invisible vehicle, since these two clusters are close to each other.

With the extended bounding box, the approximate area where the detections from the partially visible vehicle are located can be determined; thus, the merging error is reduced, as shown in Fig. 7(b) and (e). However, splitting occurs in the cluster of the invisible car on the left. By introducing the merging algorithm, splitting is reduced for the invisible cluster without introducing merging error to these two clusters, as shown in Fig. 7(c) and (f).

To numerically evaluate the performance of the proposed clustering approach on the dataset, the results are presented using the aforementioned metrics for all of the three clustering approaches in Table X.

As shown in Table X, the radar-only clustering with the proposed parameters sharply reduces the merging error compared with that without the proposed parameters, but the splitting error in its results slightly increased. The results of the camera-assisted method without the merging step show a much higher clustering accuracy and less merging error compared with those of the radar-only method. However, there are more splitting errors due to the bounding box error and the low resolution of radar. By adding the merging step, the clustering accuracy is further improved, and the splitting errors are largely reduced while no more merging errors are introduced. Compared with Table VIII, these results show the effect of promoting better clustering under low-resolution conditions of our proposed method.

The performance evaluation, conducted using the nuScenes dataset, is presented in Table XI. This evaluation is executed employing the *timeit* module in Python on a laptop equipped with an Intel Core i5-12500H CPU and 16 GB of RAM. The findings indicate that the incorporation of the proposed criteria for radar-only clustering only slightly increases the average computational time, which is nearly negligible. The camera-assisted methodology takes three to four times the time radar-only clustering takes, which is still in the same order of

TABLE X
PERFORMANCE EVALUATIONS ON NUSCENES DATASET

Metric	Radar-only Without proposed parameters	Radar-only With proposed parameters	Camera-assisted Without merging	Camera-assisted With merging
Correct clustering (%)	85.08	93.58	97.29	99.13
Average number of splitting	0.05	0.2	0.45	0.1
Average number of merging	2.2	0.45	0.1	0.1

TABLE XI
RUNTIME COMPLEXITY ON NUSCENES DATASET

Metric	Radar-only Without proposed parameters	Radar-only With proposed parameters	Camera-assisted Without merging	Camera-assisted With merging
Average runtime per frame (ms)	5.3	5.4	16.2	21.6
Average runtime per measurement (ms)	0.19	0.20	0.59	0.78

TABLE XII
PERFORMANCE EVALUATIONS ON BOREAS DATASET

Metric	Radar-only Without proposed parameters	Radar-only With proposed parameters	Camera-assisted Without merging	Camera-assisted With merging
Correct clustering (%)	18.33	61.18	77.56	88.78
Average number of splitting	0.05	1.0	2.625	1.25
Average number of merging	22.69	2.38	0.06	0.06

magnitude compared with that of radar-only clustering. The computational efficiency may be improved if the proposed method is implemented in a different programming language or if the aforementioned functions are substituted with more optimized ones.

2) Experimental Results on Boreas Dataset: To ascertain the generalizability of the proposed methodology across diverse datasets, a subset comprising 16 data frames, each featuring manually identifiable radar detection clusters, was extracted from the scenario 2021-09-02-11-42 of the Boreas dataset for this analysis. The parameters chosen for analysis are consistent with those presented in Tables II and IX.

Only Lidar detections are provided in the Boreas dataset, and they are downsampled by 20 times each in range and azimuth direction to mimic the density of radar detections. However, there are still lots of clutter points from the background in the prepocessed data. Meanwhile, some of the Lidar frames are not synchronized with camera frames well and the registration error is not negligible, which will also reduce the clustering accuracy. The evaluation results on the Boreas dataset are shown in Table XII.

As shown in Table XII, the radar-only clustering without the proposed parameters suffers from severe merging errors in these unclean frames with large amount of noise points. With the proposed criteria, the accuracy of radar-only clustering sharply increases. Our proposed camera-assisted clustering method further increases the clustering accuracy by utilizing bounding box information, and the merging step reduces the splitting errors due to the inaccurate object detection with images and registration errors. This result shows the applicability of different datasets of our proposed method.

VIII. CONCLUSION

This article addresses the challenges of radar measurement clustering in autonomous driving, with an emphasis on improving clustering through the incorporation of camera data. This work explores two distinct clustering methodologies: one that relies solely on radar data, and another that is augmented by camera data. For the radar-only approach, the study introduces range-adaptive criteria. In the camera-assisted approach, the bounding box ID associated with each radar measurement is employed for clustering purposes. This study also proposes class adaptive parameters specifically designed for short-range clustering, intending to minimize errors related to splitting and merging. To tackle the issue of coverage discrepancy, the bounding boxes of partially visible targets are expanded. Furthermore, a 2-D-assignment-based merging algorithm is introduced as a means to decrease splitting errors. The impact of these clustering methodologies is assessed through extensive simulation and the utilization of the nuScenes self-driving dataset. The results indicate a notable enhancement in clustering accuracy when employing the proposed camera-assisted methodology.

The proposed approach has room for improvement. Its performance depends heavily on the accuracy of camera information. The proposed algorithm mitigates the impact of unreliable visual data through clustering and merging, but it is still inferior to other algorithms when the camera provides little additional information. Consequently, the proposed algorithm is not robust to unseen adverse weather conditions. The future work will investigate the influence of camera visibility and enhance the applicability of the proposed algorithm in various weather scenarios.

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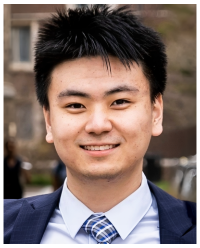

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